Water Treatment Technology (CLPE16)

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Lecture 16-17 & 18 (22, & 26 August 2025)

Discussion on:

- Introduction to physical/chemical water treatment Process

5. Overview method used for water treatment

- Discussion on Aeration
- Various Aeration Process
- Types of Aerator

- Discussion on Removal of Solids (Sedimentation or Clarification)
- Types of particles
- Types of suspension
- Classification of Settling

Water Treatment Technology (CLPE16)

Stream: **B.Tech.** Year: **IVth Year** Semester: **VIIth Sem.**
Date: **19/08/2025** **Time: 3:00 PM - 4:00 PM**

Cycle Test 01- Examination

Time Duration: 1 hour **Exam Hall: LH09** **Total Marks: 20**

Class Test Exam Instructions:
1. All questions in this paper are compulsory and write in answer booklet.
2. Make sure to read the instructions for each question before you begin answering.
3. Fill out your name, roll number and other details on the answer booklet before submission.
4. Mobile phones and smart watch are not allowed inside the exam hall.
5. Any form of cheating or misconduct will result in immediate disqualification from the exam.

Q1. In assessing wastewater, a BOD/COD ratio close to _____ indicates that most of the organic material is biodegradable. **[1]**

Q2. Write the amount of total dissolved solid (TDS) per liter for the following water: **[1]**
(a) Fresh Water _____ mg/L TDS
(b) Saline Water _____ mg/L TDS

Q3. Volatile suspended solid (VSS) is measured by igniting a sample at _____ degree Celsius. **[1]**

Q4. The _____ formula used in Incremental increase method to estimate the future population sizes. **[1]**

Q5. The _____ method involves observing how much light is scattered by suspended particles in water. **[1]**

Q6. The chemical agent typically used in COD tests is _____. **[1]**

Q7. The difference between total solid and total fixed solid gives _____. **[1]**

Q8. The _____ method uses birth, death and migration rates to estimate future population sizes. **[1]**

Q9. Calculate the Theoretical oxygen demand (ThoD) for 100 mg/L of acetic acid (CH_3COOH) and ethanol (C_2H_5OH). The stoichiometric chemical reaction for complete oxidation of acetic acid and ethanol is: (a) $CH_3COOH + 2O_2 \rightarrow 2CO_2 + 2H_2O$ (b) $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ **[2]**

Q10. In the laboratory, two 0.5-liter sludge samples (Sample A and Sample B) are analyzed. After drying at 103°C, the total solids (TS) are found to be 0.520g for Sample A and 0.270g for Sample B. The dried residues are then ignited at 550°C until only the total fixed solids (TFS) remain: 0.180g for Sample A and 0.090g for Sample B. Determine the total volatile solids (TVS) in mg/L for both samples? **[2]**

Q11. Draw and explain the general water treatment flow diagram for surface water having organic contaminated? What are the common treatment processes used for the surface water purification? Explain the direct and in-line filtration water treatment process? **[4]**

Q12. Define Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD). Explain the typical reactions involved in BOD and COD experiments, highlight the key differences between them, and discuss how BOD kinetics are mathematically represented along with the factors influencing the rate constant (k)? **[4]**

Q. How City Water Purification Works?

https://www.youtube.com/watch?v=KsVfshmK0Ak

Aeration

- It refers to the process of introducing air into liquids/substance/soil to facilitate the exchange of gases.
- In the context of water treatment, aeration involves mixing air with water to increase the oxygen content and remove unwanted gases, materials or volatile substance.
- This process is important in various applications such as water and wastewater treatment, agriculture, improve the quality and characteristic of the treated substance.

The purpose of aeration is to remove unwanted gases present in the water as well as unwanted materials in water in the presence of oxygen.

For Example:

- Unwanted gases are Hydrogen Sulfide (H_2S) and Carbon dioxide (CO_2)
- Unwanted materials are Iron (Fe) and Manganese (Mn)

Both can be removed by **Aeration process**

Q. How H_2S and CO_2 coming into water?

If water is rain water → the amount of dissolved H_2S and CO_2 in water will be very very negligible.

So, No need to remove them from water

But if water is ground water and specially if it is under high pressure → the solubility of H_2S and CO_2 is very very high in water.

- If H_2S and CO_2 present in water → It will produce undesirable smell

So, We can not supply this water.

We need to remove this undesirable smell and odor of that water.

Q. How to remove smell and odor?

We have to provide excess amount of oxygen to that water.

The water will become over saturated with oxygen. So, in this way gases will coming out. Once these gases removed the odor of the water getting removed.

- Similarly, → We can remove unwanted metals like Iron (Fe), and Manganese (Mn)

For example → If we take Iron (Fe), and Manganese (Mn)

- The Iron (Fe) will be present in the form of Fe^{2+} and Manganese (Mn) in the form of Mn^{2+}

➢ If we provide excess oxygen than;

Fe^{2+} will oxidized to Fe^{3+}

$$Fe^{2+} \rightarrow Fe^{3+}$$

Mn^{2+} will oxidized to Mn^{4+}

$$Mn^{2+} \rightarrow Mn^{4+}$$

Once, It is oxidized to Fe^{3+} and Mn^{4+} then, we can easily remove them by precipitation.

- The chemical reaction, we can write like this for Fe:

$$4Fe^{2+} + O_2 + 10H_2O \rightarrow 4Fe(OH)_3 \downarrow + 8H^+$$

- Similar for Mn, we can write like this:

$$2Mn^{2+} + O_2 + 2H_2O \rightarrow MnO_2 \downarrow + 4H^+$$

In some case like → Volatile Organic Compound (VOC) – Phenol, Benzene, Toluene, Xylene

Aeration can be used, but it can not be remove completely.

If the concentration of VOC is very high -

Then also we can go for the aeration and bring the concentration down to the hand able level and we can go for the other treatment level like adsorption or ion-exchange etc.

In short, We can tell that aeration is used for the removal of unwanted gases like H_2S and CO_2 and unwanted materials like Fe and Mn

How can we achieve aeration?

Various aeration process:

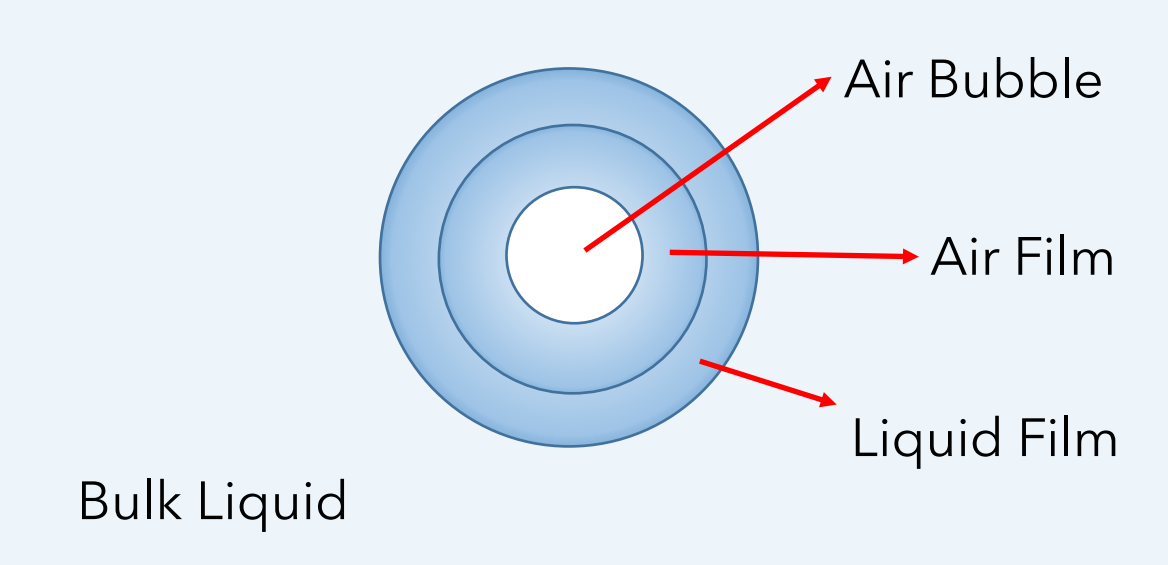
Aeration can be done either:

(a) Dispersing air into water, and

(b) Dispersing water into air

Both way we can carry aeration

(a) Dispersing air into water: We are injecting high pressure air into the water. Which has to be aerated. As shown in the figure, at the center the air bubble, then air film and liquid film and the bulk liquid. So, we are dispersing the air into the liquid.



The air bubble is there into the bulk liquid and air bubble have a air film surrounded to that one. After the air film there is a liquid film and the bulk liquid.

Let's assume, $\mathbf{C_{PBL}}$ is the concentration of pollutant in the bulk liquid and $\mathbf{C_{PAB}}$ is the concentration of pollutant in the air bubble.

- **If $C_{PAB} < C_{PBL}$ →**

  mean's the concentration gradient will be form bulk liquid to air bubble. The pollutant will moving towards air bubble side. This is known as desorption.

- **If $C_{PAB} > C_{PBL}$ →**

  The concentration gradient will be towards bulk liquid side. Mean's oxygen will be transferred from the air bubble to the bulk liquid. This is known as adsorption.

Various aeration process:

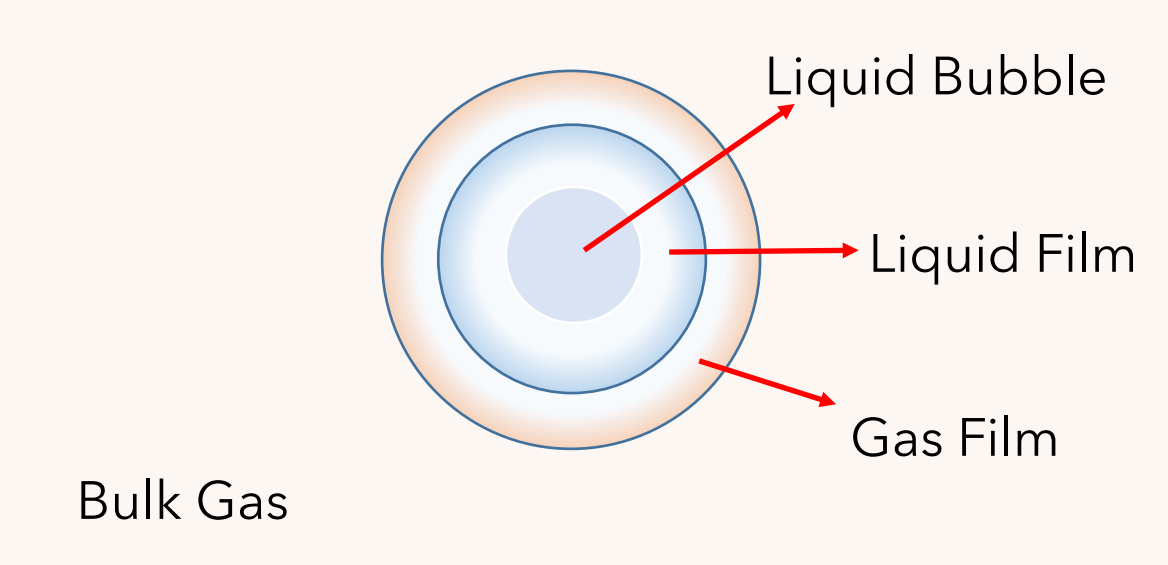
Aeration can be done either:

(a) Dispersing air into water, and

(b) Dispersing water into air

Both way we can carry aeration

(b) Dispersing water into air: The liquid bubble at the center, then liquid film and liquid film having gas film.



Let's assume, C_{PBG} is the concentration of pollutant in the bulk gas and C_{PLB} is the concentration of pollutant in the liquid bubble.

- **If $C_{PLB} < C_{PBG}$ →**

  Naturally, the pollutant will be moving from liquid to the gas. This is known as desorption.

- **If $C_{PLB} > C_{PBG}$ →**

  Naturally, the pollutant will be moving from gas to liquid side. This is an example of adsorption.

So, this is the working principle of Aeration.

In case of waste water treatment system; in the aeration → We usually go for Dispersing water into air.

Types of aerators:

The different aerators adopted for dispersing water into air. Most commonly found is:

(a) Fountains/Diffuse Aerator

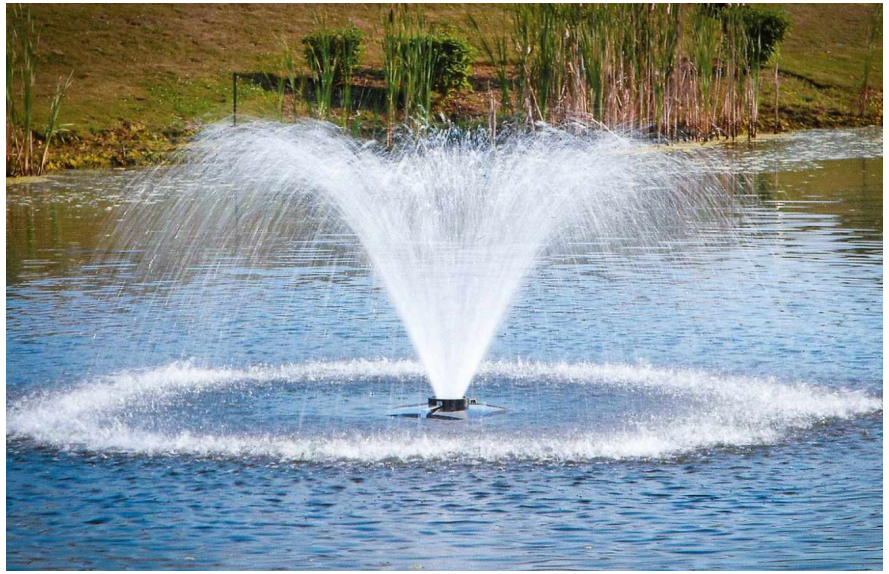
(b) Cascade Aerator

(c) Tray Aerator

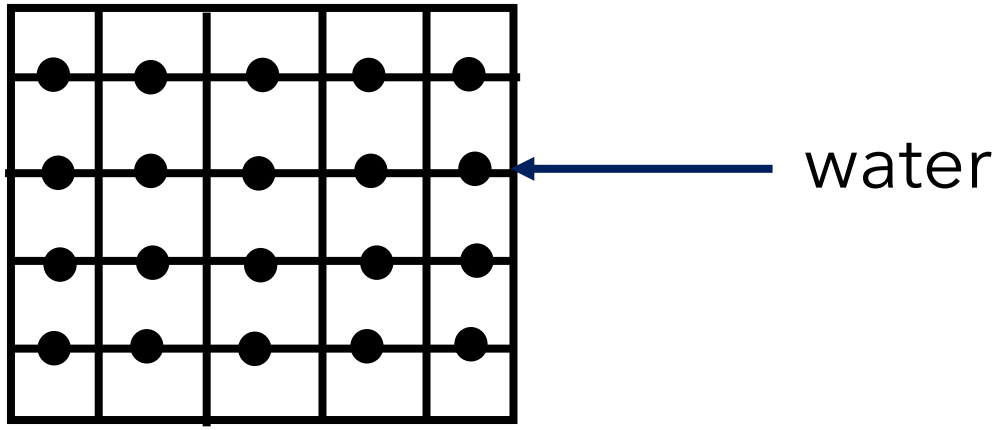
(a) Fountains/Diffuse Aerator:

In fountain aerators are a grid of pipeline provided above a basin. This grids of pipeline having a nozzle at various location. Nozzles are provided in such a way that the air will be coming out under high pressure in the upward direction. Water is coming out under high velocity.

- Nozzle size varies from → 2-4cm
- pressure is maintained around → 70 kpa
- Area required for this system ~ 10 m2/50L/sec
- Space between the pipe ~ 6–350 cm.



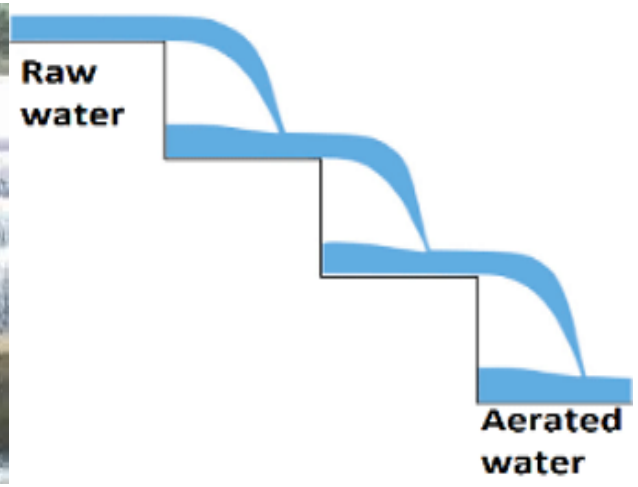
- The water having high kinetic energy. Once the water loses this kinetic energy, water will be falling down and it will be collected in the bottom of basin. So, during that time it is coming into the contact with atmosphere and naturally the contact area is vary high and lots of gas transfer will be taking place. During that time whatever ordo gas present in the water will escaping and similarly whatever metal present in water. It will oxidized and can be removed easily.

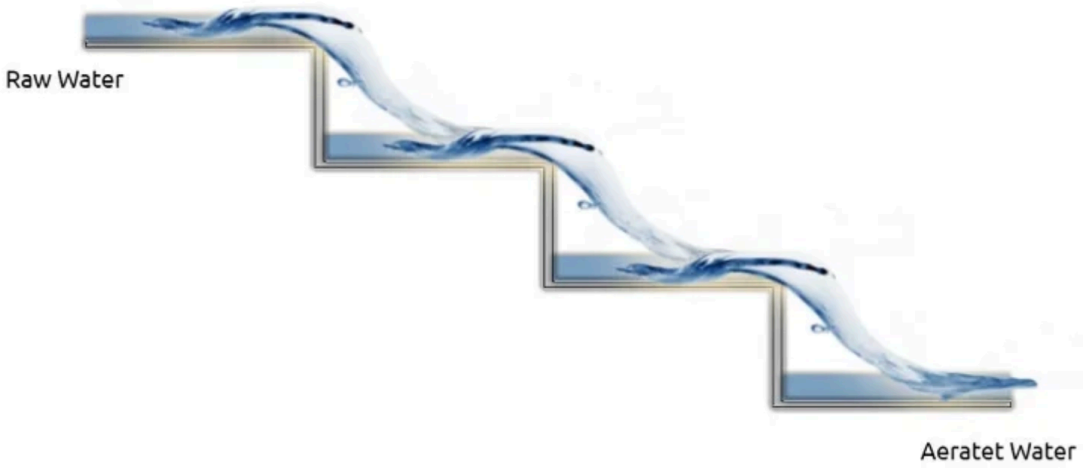


(b) Cascade Aerator:

Water is lifting to the top of the cascade and allow to pass through this and water will be falling like step by step by step and it is collecting at the bottom. Most of the time the step height will be around 30 cm and number of steps will be 10.

Area required for this system is 4-9 m^2/50L/sec

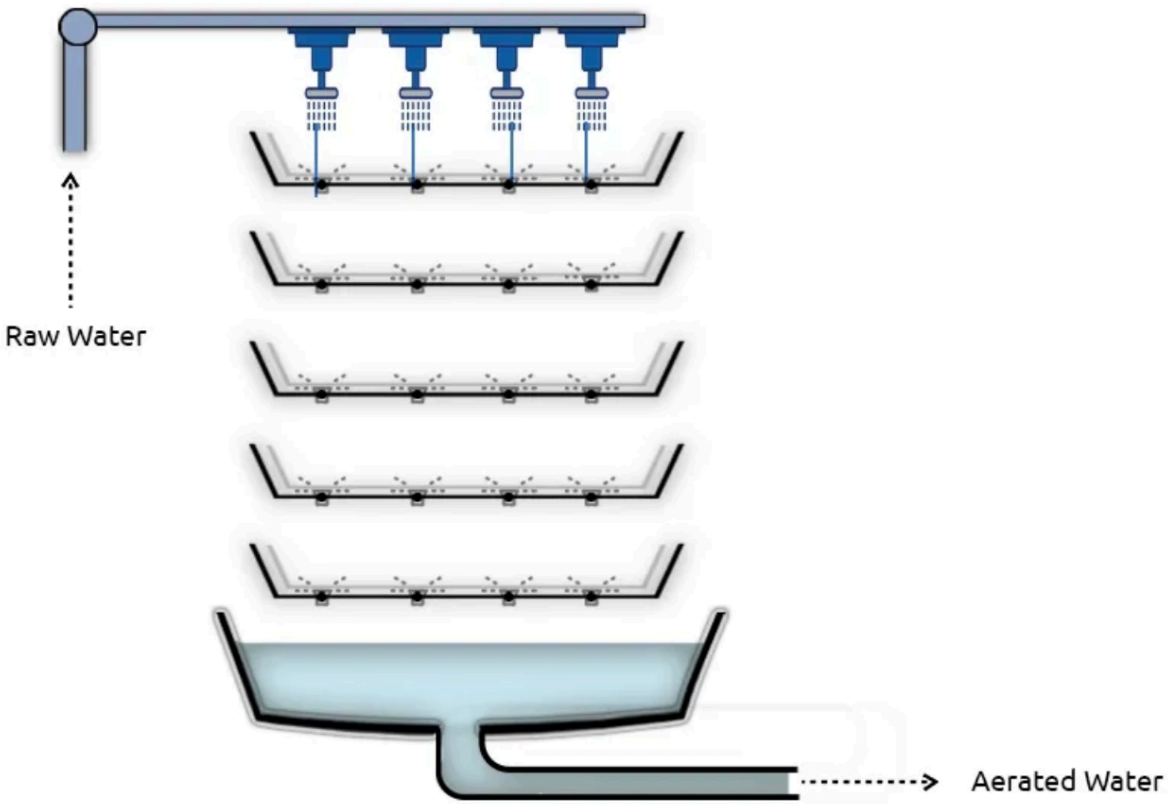




What is head loss? It is the total height from the top to the bottom (till water collection system)

(c) Tray or multiple tray Aerator:

It is similar to cascade aerator only difference is the water is allow to pass through solid media and reaction will be taking place on the solid surface. So more contact will be there and reaction will be much faster.



Removal of Solids

- Sedimentation or clarification unit used to remove the solids.

Solids →
- Suspended solids
- Colloidal Solids

Q. What is happening in sedimentation tank?

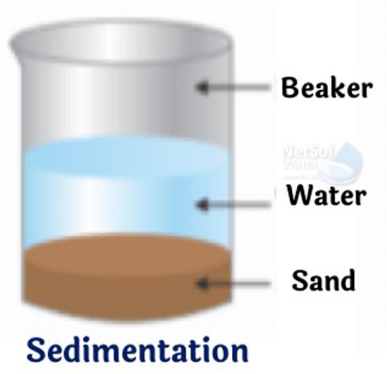
Because of the gravitational force and weight of the solid particles

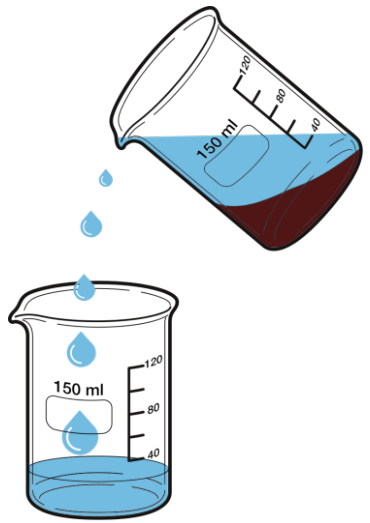
→ The particle will be settling down

Q. What we have do in sedimentation tank?

- Allow the water to stay quiescent in some settling basin or sedimentation tank and give enough detention time

- After some period of time, the particle will be settling down and we can collect the clean water from outlet.





This settling is depending upon the liquid as well as particle characteristic.

Particles can be divided in two parts:

(1) Discrete Particles

(2) Flocculent Particles

(1) Discrete Particles

- Discrete particles are the one which do no change its size, shape and specific gravity or the properties with respect to time, those are known as **discrete particles**.
- That means the particles will be settling as independent particles. It will not affected by the surrounding particles.

(2) Flocculent Particles

- Flocculent particles are the one which change its size, shape and specific gravity or the properties with respect to time, those are known as **flocculent particles**.
- The settling velocity will not be a constant. We can not find the settling velocity.

Because the particles keep on changing its size, shape, and specific gravity with respect to time.

Only by conducting experiment – we can find the settling velocity

THANK YOU
Do you have Any Question?